

U-ESFM: publication scenarios

Three outcome-conditional plans (A / B / C), revised to incorporate the RESfM table extraction, Fadi Khatib's metric priority, Ronen Basri's point on label cost, and the reproduction gap. Dated 11 August 2026.

Where things actually stand

Two facts govern everything below, and both are unresolved. Until they are, no scenario can be selected — they are not refinements of the plan, they determine which plan exists.

1. The reproduction gap (blocking)

Our **RESfM (released)** row — the authors' official pretrained checkpoint, our protocol, **our tracks** — sits at $10.59^\circ / 10.21^\circ$ mean rotation on 1DSfM. Their published Table 2 averages **3.98°** over the same ten scenes. That is a $2.6\times$ gap, and it is the difference between winning and losing: U-ESFM is at $6.93^\circ / 5.73^\circ$, comfortably ahead of 10.2° and comfortably behind 3.98° .

The checkpoint is theirs, so retraining is not a confound. **The tracks are the prime suspect.** Their outlier head was fitted to their track construction and its 0.6 removal threshold was tuned on that score distribution; different tracks shift the distribution under a fixed threshold. This is a benign explanation, not a bug — but it must be demonstrated, not assumed. Scene subsetting is unlikely to explain it: eight of their ten per-scene values lie between 0.67° and 3.96° .

2. The metric priority (blocking)

Fadi's ordering is **translation** → **rotation** → **registered cameras**. Every crossover number we have is **rotation**. Under his ordering we are claiming victory on the secondary metric, and whether translation also crosses is unknown from results already computed. Note this ordering is **verbal, not published** — RESfM's Section 4.3 lists both metrics without ranking them, so it can be adopted as our convention but cannot be cited to them.

Do first, this week, no new training runs: (i) per-scene outlier percentage of our tracks vs their published values, for all ten 1DSfM scenes; (ii) translation error for every method at every protocol tag; (iii) reconcile 5.73° vs 6.93° — which protocol, which loss configuration, which seeds. All three are queries against existing results, and together they select the scenario.

What the RESfM tables changed

- **The label-cost argument is dead, as Ronen said.** COLMAP labels are cheap, so removing supervision is a design choice, not a contribution. The paper must be an empirical claim about *where the supervision is valid*.
- **The in-distribution contamination story is not supported by their data.** RESfM's error does not track contamination on MegaDepth (Spearman $\rho = +0.29$, $p = 0.086$); its best band is 25–30% outliers at 0.19° . If we want that mechanism we must produce it ourselves via controlled injection.
- **The shift story is supported.** MegaDepth → 1DSfM degradation: RESfM $\times 3.10$, GASFM $\times 2.94$, Theia $\times 2.39$, ESfM $\times 1.65$, GLOMAP $\times 0.51$. The supervised head degrades *more* than the unsupervised baseline it improves on.
- **Their own classifier is weak under shift.** Table 7 on 1DSfM: mean inlier recall 56.0%, outlier F-score 54.1, contamination reduced only 31.9% → 19.0%. Tower of London collapses to 28.3% inlier recall. Quote inlier recall, never precision — precision rises mechanically with prevalence.
- **GLOMAP is a threat only on rotation.** On 1DSfM it beats RESfM 1.27° vs 3.98° (7/10 scenes) at 98% coverage vs 72% — but loses translation, 0.576 vs 0.427. Under Fadi's ordering RESfM wins; under a

rotation-led claim we lose to a training-free classical method.

- **1DSfM confounds shift with contamination.** Its scenes run 25.1–39.4% outliers (mean 31.9) against MegaDepth's ~25%. A 1DSfM win alone cannot separate the two hypotheses.
- **Training-set contamination is never reported.** Only test scenes. The best proxy is the 13 Group-1 test scenes: mean 29%, median 30%, up to 44.6% — so we cannot argue the head never saw dirty scenes.
- **ESfM's coverage convention is available to us.** Moran et al. parenthesise every result where $\geq 10\%$ of cameras are unregistered and treat full coverage as a selling point. Adopting it is continuity, not special pleading — and under it, most of RESfM's 1DSfM row would be marked.

Scenario A — the shift claim holds on translation

Trigger

Reproduction gap resolved (or shown to be a track-distribution effect, with RESfM given its best threshold on our tracks), **and** U-ESfM beats the properly-calibrated RESfM on 1DSfM **translation**, not only rotation.

The claim

Not “we remove the need for labels.” The claim is that RESfM's outlier head is fitted to one domain's contamination statistics and must generalise zero-shot, while U-ESfM's percentile mechanism recalibrates per scene by construction. Under distribution shift — the deployment condition, since nobody reconstructs held-out MegaDepth — the self-supervised mechanism wins. Label cost is a footnote.

Required experiments

- **Three-row RESfM decomposition:** published / released-on-their-tracks / released-on-our-tracks. Row 2 isolates our eval path, row 3 isolates the track effect. Their preprocessed tracks were to be released publicly — check, and ask Fadi if not.
- **Steelman: RESfM's threshold swept on our track distribution.** Comparing our adaptive mechanism against their *miscalibrated* fixed threshold is not a comparison, and is the first thing a reviewer will ask us to redo.
- **Steelman: RESfM retrained on target-domain COLMAP labels.** If labels are cheap, the obvious counter is “regenerate them on 1DSfM and retrain the head.” Run it. If we still win, the claim is unqualified; if not, it narrows honestly to *zero-shot* — no target reconstruction, no retraining — which is still true and still publishable.
- **Classical column in every main table**, GLOMAP above all, at matched coverage.
- **Contamination injection sweep** (5/10/20/30/40/50% on fixed MegaDepth scenes) to separate shift from contamination, and to measure COLMAP label error against injected ground truth.
- **Strecha + BlendedMVS** with a single fixed configuration — see the role note below.

The role of Strecha and BlendedMVS

They are saturated (0.027° and 0.011° mean rotation; RESfM ranks 3rd and 2nd behind Theia and COLMAP) and $n=4$ each, so they cannot be a headline. They do four things 1DSfM cannot: **(1)** their ground truth is LIDAR and rendered, *not* COLMAP, which is our only answer to the circularity objection; **(2)** they are the only tables where MAST3R and VGGSfM appear, so reporting them inherits the feed-forward comparison reviewers will demand; **(3)** they are exactly where RESfM hand-tunes its threshold from 0.6 to 0.8, so matching there with one fixed configuration demonstrates the tuning argument instead of asserting it; **(4)** they anchor the low-contamination end of the axis (2.7% and 3.6%). Present them as a *no-collapse plus validity* result, not a second win.

Objections to preempt

- **Circularity.** MegaDepth GT is COLMAP output, so “COLMAP labels are noisy” seems to indict our own metric. Answer explicitly: pose estimation aggregates thousands of observations; RESfM's labels are per-observation classifications from track matching plus a 4px threshold (their Appendix C). Aggregation tolerates noise that per-item labelling does not.
- **“You crippled the baseline.”** If the gap is a track-distribution effect, that *is* a version of our thesis — but stated carelessly it reads as feeding RESfM inputs it was not built for while our mechanism adapts freely. Report both track sets, always.
- **GLOMAP improves on 1DSfM** ($\times 0.51$). So 1DSfM is not simply “harder” — frame the shift as distributional, never as difficulty.
- **“Just use RESfM where it was trained.”** A deployed system does not know whether a new collection matches the training contamination profile. Put this in the abstract, not the discussion.

Venue and framing

ICLR 2027 — abstract 18 Sep, paper 25 Sep 2026. A learning-objective story, RESfM's own venue, and a conditional claim reads there as a contribution rather than a missing SOTA number. Drop TTT from the headline if its OOD contribution remains nil; make the claim about zero-shot transfer of the self-supervised mechanism and demote TTT to in-distribution analysis. Two figures carry it: the ID-vs-OOD Pareto plot (we are *elsewhere on the frontier*, not behind) and per-method ID→OOD degradation bars.

Scenario B — the win is real but narrow

Trigger

Any of: crossover holds on rotation but not translation; holds only on 1DSfM and vanishes elsewhere; sits inside seed noise; or survives only against RESfM-on-our-tracks and not against the swept-threshold or retrained steelman.

The claim

A solid unsupervised alternative that is honest about the in-distribution gap and degrades more gracefully under shift. Scoped explicitly to *deep* methods if we do not beat GLOMAP on translation: “among learned SfM methods...” — with the classical column still reported in the main table, never omitted.

What changes from A

- Drop the contamination-injection sweep from the critical path; keep it as an appendix if it runs.
- Keep both steelman baselines — they are what make a narrow claim credible rather than under-tested.
- Lead with the mechanism figures (Table 7 inlier recall, ID→OOD degradation ratios) rather than with a headline win, since the mechanism evidence is stronger than the margin.
- State the in-distribution tax plainly in the abstract. A Pareto framing is honest here; a SOTA framing is not.

Venue

3DV 2027 was 28 Aug 2026 — realistically gone given the reproduction gap. The next geometry-native window is **ICCV 2027 (~March 2027)**. ICLR remains possible if the desk checks resolve quickly and cleanly, but a narrow claim at ICLR is a harder sell than the same claim at a 3D venue.

Scenario C — no crossover survives

Trigger

The gap is an artifact: properly-calibrated RESfM beats U-ESFM on both metrics under shift; or the statistical two-stage MAD baseline matches the learned mechanism, making the

supervised-vs-self-supervised framing moot.

What to do

Shelve the main-venue ambition rather than force it. “Unsupervised, slightly worse, adaptation doesn't help” has no main-venue home — and RESfM's labels come free from COLMAP, so the eliminated supervision was never expensive. This is a legitimate outcome, not a failure: a clean negative result on whether the learned outlier head earns its place over robust BA plus a statistical rule is genuinely useful to this subfield.

Workshop options: CVPR 2027 workshops (~March 2027) or a NeurIPS 2026 workshop this autumn. Note 3DMV and IMW are **archival**, so a workshop paper forecloses the same content elsewhere. If targeting IMW, Mishkin's earlier feedback still stands: he wanted the IMC benchmark and fundamental-matrix filtering, neither of which was added.

Decision table

	A	B	C
Translation crossover on 1DSfM	yes	no (rotation only)	no
Survives swept-threshold steelman	yes	yes	no
Survives retrained-head steelman	yes, or narrows to zero-shot	narrows	no
Beats best classical on translation	yes	not required	no
Venue	ICLR 2027 (25 Sep)	ICCV 2027 (~Mar)	Workshop
Injection sweep on critical path	yes	no	no

Calendar and the one-shot problem

ICLR (25 Sep, decisions 16 Dec) and CVPR 2027 (~13 Nov) **overlap**: an ICLR rejection lands after the CVPR deadline. These are mutually exclusive one-shot options this autumn, not a sequence. 3DV 2027 (28 Aug) is effectively gone. After a December rejection the next window is ICCV 2027 in roughly March; ECCV is even-year, so ECCV 2028.

Requirements common to all three scenarios

- **Report translation first**, mean and median. RESfM's 1DSfM translation is 0.427 mean against 0.078 median — the two can rank methods differently.
 - **Report Nr in every table**, and adopt ESfM's convention of parenthesising results with $\geq 10\%$ of cameras unregistered. Where coverage diverges materially, also report errors on the shared camera intersection.
 - **One fixed configuration across all datasets**. We intend to argue that per-domain threshold tuning is a weakness; we must not do it ourselves. This is a hard constraint on the Strecha/BlendedMVS runs.
 - **Fixed row naming**: RESfM (published) / RESfM (released, their tracks) / RESfM (released, our tracks). Never collapse them into one column.
 - **Naming**: U-ESfM means the adaptive-loss method only. The statistical variant is “two-stage MAD” and is a baseline. Every U-ESfM number carries a protocol tag.
 - **≥ 3 seeds per cell** where compute allows; single-seed cells marked $n=1$ and cannot support a claim.
 - **Decide explicitly whether reprojection error appears**. ESfM led with it; RESfM dropped it. Either is defensible; silently doing neither is not.
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Open questions for Fadi are in the companion document. The highest-value asks given the above: their preprocessed point tracks, their trained ESFM/GASFM weights, GLOMAP versions and settings, and whether the metric priority is lexicographic or weighted.